📄 Crypto Watcher System – Technical Documentation

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1. Introduction

The Crypto Watcher system is an automated cryptocurrency monitoring solution built using **Make.com**. It is designed to track real-time cryptocurrency market data, apply filtering logic, and deliver alerts when significant market movements occur.

To enhance usability, a frontend dashboard was also developed using Lovable, providing a visual interface for monitoring crypto metrics and alerts.

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2. Objectives

The system was developed to achieve the following:

- Monitor cryptocurrency prices and related metrics automatically
- Trigger workflows via webhook
- Fetch real-time crypto data from a public API
- Apply logic to detect significant price changes
- Send real-time alerts to users
- Log data for tracking and analysis
- (Bonus) Provide a responsive UI dashboard for visualization

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3. System Architecture

The system follows a simple automation pipeline:

Webhook → HTTP Request → Iterator → Filter → Telegram → Google Sheets → Lovable UI

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4. Tools and Technologies Used

- **Make.com** – Workflow automation platform
- **CoinGecko API** – Source of cryptocurrency data
- **Telegram Bot** – Notification delivery
- **Google Sheets** – Data logging and tracking
- Lovable – Frontend dashboard generation

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5. Implementation Details

5.1 Webhook Trigger

A custom webhook was created in Make.com to initiate the workflow.
This allows the system to be triggered externally (e.g., via browser or scheduler).

5.2 Crypto Data Retrieval

An HTTP module was used to fetch live cryptocurrency data from the CoinGecko API.

API Endpoint Used:

https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd

Data Retrieved Includes:

- Coin name
- Current price (USD)
- 24-hour price change (%)
- Market capitalization
- 24-hour trading volume
- Last updated timestamp

5.3 Data Processing (Iterator)

The Iterator module was used to process the API response.

Since the API returns multiple coins in a list, the iterator breaks the data into individual items, allowing each coin to be evaluated separately.

5.4 Logic and Filtering

A filter was applied to identify significant market movements.

Condition used:

- Price change > 5%
  OR
- Price change < -5%

This ensures alerts are triggered for both upward and downward movements.

5.5 Notification Delivery (Telegram)

A Telegram Bot was integrated to send real-time alerts.

Alert Message Includes:

- Coin name
- Current price
- Percentage change
- Market capitalization
- Trading volume
- Timestamp

This enables users to quickly react to market changes.

5.6 Data Logging (Google Sheets)

Google Sheets was used to log filtered results.

Logged Data:

- Name
- Price
- Change (%)
- Market Cap
- Volume
- Time

This allows for:

- Historical tracking
- Trend analysis
- Record keeping

5.7 Lovable UI (Bonus Feature)

A frontend dashboard was created using Lovable.

The dashboard connects to the Make.com webhook and displays crypto data in a user-friendly format.

Features:

- Live crypto data display
- Alert section for significant changes
- Responsive design (mobile + desktop)
- Clean and interactive layout

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6. Testing and Validation

The system was tested end-to-end to ensure proper functionality.

Test Results:

- ✅ Webhook successfully triggers the workflow
- ✅ API retrieves accurate and complete data
- ✅ Iterator processes multiple coins correctly
- ✅ Filter correctly identifies ±5% price changes
- ✅ Telegram alerts are delivered in real-time
- ✅ Google Sheets logs data correctly
- ✅ Lovable UI updates dynamically when triggered

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7. Challenges and Solutions

Challenge 1: Understanding Data Structure

- The API returned data in an array format
- **Solution:** Used Iterator to process each item individually

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Challenge 2: Filtering Logic

- Needed to capture both positive and negative changes
- **Solution:** Applied OR condition (>5% OR < -5%)

Challenge 3: UI Integration

- Connecting frontend to backend without coding
- **Solution:** Used Lovable to generate and connect dashboard via webhook

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8. Conclusion

The Crypto Watcher system successfully meets all the requirements outlined in the PRD.

It provides:

- Automated crypto monitoring
- Intelligent filtering
- Real-time notifications
- Data logging
- Interactive dashboard visualization

The system is scalable and can be extended with additional features such as multi-channel alerts, advanced analytics, or portfolio tracking.

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9. Future Improvements

- Add support for multiple notification channels (Slack, Email, SMS)
- Implement historical charts and trend analysis
- Allow users to track selected coins
- Enhance UI with interactive charts
- Integrate scheduling for automatic periodic execution